norms

$Q^* = Q^{-1} \Leftrightarrow QQ^* = Q^*Q = I \Leftrightarrow$ cols orthonormal

unitary preserve: $(Qx)^*(Qy) = x^*y$

$$\|Qx\| = \|x\|$$

Cauchy Sw: $\langle x,y \rangle \le \|x\|\|y\|$

$$\|A\|_{(m,n)} := \sup_{\|x\|_n = 1} \|Ax\|_m$$

$$\|AB\|_{(l,n)} \le \|A\|_{l,m}\|B\|_{m,n}$$

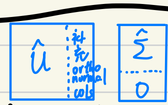
$\|A\|_1 = \max_{1\le j\le n} \|a_j\|_1$ (= max abs col sum), $\|A\|_2 = \sigma_1$

$\|A\|_\infty$ = max abs row sum

$$\|A\|_F = \sqrt{\sum\sum |a_{ij}|^2} = \sqrt{\sum_n \|\vec{a_j}\|^2} = \sqrt{tr(A^*A)} = \sqrt{tr(AA^*)} = \sqrt{\sum_1^r \sigma_i^2}$$

$\|AB\|_F^2 \le \|A\|_F^2 \|B\|_F^2$, $\|QA\|_2 = \|A\|_2$, $\|QA\|_F = \|A\|_F$

SVD: $v_i \mapsto \sigma_i u_i$



reduced: $A = \underset{\substack{m\times n\\ \text{(ortho cols)}}}{\hat{U}} \underset{\substack{n\times n\\ \text{diag}}}{\hat{\Sigma}} \underset{\substack{n\times n\\ \text{(ortho cols)}}}{V^*}$ full: $A = \underset{\substack{m\times m\\ \text{uni}}}{U} \underset{\substack{m\times n\\ \text{diag}}}{\Sigma} \underset{\substack{n\times n\\ \text{uni}}}{V^*}$

solve $Ax=b$ by SVD: $A = U\Sigma V^*$, $x' := V^*x$, $b' := U^*b \Rightarrow b' = \Sigma x'$

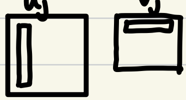
1. $\mathrm{rank}(A) = \langle u_1, \dots, u_r \rangle$, $\mathrm{null}(A) = \langle v_{r+1}, \dots, v_n \rangle$

2. nonzero $\sigma(A)$ is $\sqrt{\lambda(A^*A)}$

3. $\det A = \prod \sigma_i$

4. $A = \sum_{j=1}^{r} \sigma_j u_j v_j^*$

u_j v_j^*



$\sum^{v} \sigma_j u_j v_j^*$

$\|A - A_v\|_2 = \sigma_{v+1}$

$\|A - A_v\|_F = \sqrt{\sigma_{v+1}^2 + \dots + \sigma_n^2}$

SVD 1

SVD 2

计算 SVD：

1. 求解 A^*A (or AA^*) 的 eigenvalues, $\sigma_i = \sqrt{\lambda_i}$
2. A^*A 的 eigenvectors (unit), $V = [\vec{v}_1 \cdots \vec{v}_n]$ (可能要补全)
3. $Av_i = \sigma_i u_i$, 求解 $U = [\vec{u}_1, \dots, \vec{u}_n]$
4. $A = U\Sigma V^*$

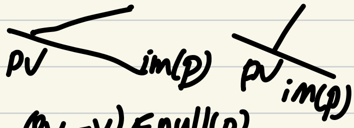
eigenvalue decomp: $A \in \mathbb{C}^{m\times m}$, $A = X\Lambda X^{-1}$, X 为 eigenvectors
A diagalizable (GM = AM)

$$\Updownarrow$$

$(A-\lambda I)x = 0$

projector

projector P : $P^2 = P$ ($\Rightarrow P^n = P$)



P projector $\Rightarrow I-P$ also projector

$v \xrightarrow[S_1]{\text{along}} S_2$ $\quad$ $v \xrightarrow[S_2]{\text{along}} S_1$

$(Pv - v) \in null(P)$

P orthogonal projector $\Leftrightarrow S_1 \perp S_2 \Leftrightarrow P = P^*$

P orthogonal projector $\Rightarrow I-P$ also orthogonal projector

orthogonal proj $P \Rightarrow \boxed{P = Q\Sigma Q^*}$ both SVD and eigendecomp

$\{q_1, \dots, q_n\}$ span S_1, $\{q_{n+1}, \dots, q_m\}$ span S_2

$\Sigma : \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix}$

can just take $[q_1, \dots, q_n]$ as $\hat{Q} \Rightarrow \boxed{P = \hat{Q}\hat{Q}^*}$ (orthogonal proj onto $im(\hat{Q})$)

specially, proj onto line of $\langle a \rangle$: $P_{\langle a \rangle} = \dfrac{aa^*}{a^*a}$

Given arbitrary basis $\{a_1, \dots, a_n\}$, $P_{\langle a_1, \dots, a_n \rangle} = A(A^*A)^{-1}A^*$

projector

QR

$A = \underset{m\times m}{Q}\,\underset{m\times n}{R} = \underset{m\times n}{\hat{Q}}\,\underset{n\times n}{R}$ $\quad$ $\vec{a_n} = \sum_{j=1}^{n} r_{jn} \vec{q_n}$

QR def

Classical GS:

$v_j \mathrel{-}= \sum_{i=1}^{j-1} q_i q_i^* a_j$

i.e. $v_j = \underbrace{(I - \hat{Q}_{j-1}\hat{Q}_{j-1}^*)}_{:= P_j} a_j$

$P_1 = I$

$O(n)$ for $j=1$ to n
 $v_j = a_j$
 $O(mn)$ for $i=1$ to $j-1$
  $\underline{r_{ij} = q_i^* a_j}$
  $v_j \mathrel{-}= r_{ij} q_i$ $\quad (-\mathrm{proj}_{q_i} a_j)$

 $r_{jj} = \|v_j\|_2$
 $q_j = v_j / r_{jj}$ (normalize) $\quad q_j = \frac{v_j}{\|v_j\|}$

$v_j \mathrel{-}= \mathrm{proj}_{\langle q_1, \dots, q_{j-1}\rangle} v_j$

Thm: $A \in \mathbb{C}^{m \times n}$ full rank $\Rightarrow$ unique reduced QR

ex. (classical GS)

$v_2 \mathrel{-}= (q_1^* a_2) q_1$

$$\underbrace{\begin{bmatrix} 1 & -1 \\ 1 & 5 \\ 0 & 6 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & 1 \end{bmatrix}}_{\text{normalize } v_1} \begin{bmatrix} 1 & -\frac{2}{\sqrt{2}} \\ 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{\sqrt{54}} \end{bmatrix}}_{\text{normalize } v_2} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{2}{\sqrt{6}} \end{bmatrix}}_{\hat{Q}}$$

$q_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}$

$$\hat{R} = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{54} \end{bmatrix} \begin{bmatrix} 1 & \frac{+2}{\sqrt{2}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & 0 \\ 0 & 1 \end{bmatrix}$$

Modified GS:

for $i=1$ to n
 $v_i = a_i$
for $i=1$ to n
 $r_{ii} = \|v_i\|$
 $q_i = v_i / r_{ii}$
 for $j=i+1$ to n
  $r_{ij} = q_i^* v_j$
  $v_j = v_j - r_{ij} q_i$

work: $\underline{\sim 2mn^2}$ flops

matrix form:

$$A \underbrace{R_1 R_2 \dots R_n}_{\hat{R}^{-1}} = \hat{Q}$$

$$R_1 = \begin{bmatrix} \frac{1}{r_{11}} & \frac{-r_{12}}{r_{11}} & \frac{-r_{13}}{r_{11}} & \cdots \\ & 1 & & \\ & & 1 & \\ & & & \ddots \end{bmatrix}$$

$$R_2 = \begin{bmatrix} 1 & & & \\ & \frac{1}{r_{22}} & \frac{-r_{23}}{r_{22}} & \cdots \\ & & 1 & \\ & & & 1 \end{bmatrix}$$

To get full QR: find $b_3 \perp \mathrm{im}(A)$, normalize it

$Q = \begin{bmatrix} \hat{Q} & \frac{b_3}{\|b_3\|} \end{bmatrix} \quad \hat{R} = \begin{bmatrix} R \\ 0 \end{bmatrix}$

Householder

for $k=1$ to n $\quad 2mn^2-\frac{2}{3}n^3$

$x = A_{k:m,k}$

$v_k = \text{sign}(x_1)\|x\|_2 e_1 + x$

$v_k = \frac{v_k}{\|v_k\|_2}$

$A_{k:m, k:n} = A_{k:m, k:n} - 2v_k(v_k^* A_{k:m, k:n})$

implicit calculate Q^*b

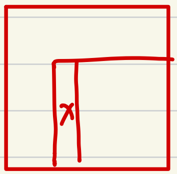
for $k=1$ to n

$b_{k:m} = b_{k:m} - 2v_k(v_k^* b_{k:m})$

implicit calculate Qx

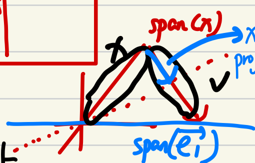
for $k=n$ to 1

$x_{k:m} = x_{k:m} - 2v_k(v_k^* x_{k:m})$



$Q_k = \begin{bmatrix} I_{k-1} & \\ & F \end{bmatrix}$

即 Q_k 把 span(x) $=$ span$(A_{k:m,k})$ reflect 到 span(e_1) 上



$F = I - 2\frac{vv^*}{\|v\|}$

$\underbrace{Q_n \cdots Q_2 Q_1}_{=Q^*} A = R$

Least squares

$\min\|r\|_2 \Leftrightarrow A^*r=0 \Leftrightarrow A^*Ax=A^*b \Leftrightarrow \text{Proj}_A b = Ax$

① use SVD to solve least sq.

$A^+ = (A^*A)^{-1}A^*$

$A = \hat{U}\hat{\Sigma}V^*$ $\quad$ $2mn^2+11n^3$

note: $\text{im}(A) = \text{im}(\hat{U})$, and U has orthonormal cols

$\Rightarrow Ax_{op} = UU^*b$ (or $\hat{U}\hat{U}^*$)

$x_{op} = A^+b = V\Sigma^+U^*b$

pesudoinverse 取 $\frac{1}{\sigma_i}$ $\forall i$ 即可

$2mn^2-\frac{2}{3}n^3$

use QR of A to solve least sq:

$\text{proj}_A b = \hat{Q}\hat{Q}^*b = Ax \Rightarrow \hat{R}x = \hat{Q}^*b \Rightarrow x = \hat{R}^+\hat{Q}^*b$

FFT

$$\tilde{F}_{2n} = \begin{bmatrix} F_n & T_n F_n \\ F_n & -T_n F_n \end{bmatrix} \qquad w_{2n} = e^{\frac{-2\pi i}{2n}}$$

$$T_n = \text{diag}(1, w_{2n}^1, w_{2n}^2, \dots, w_{2n}^{n-1})$$

F_{2n} = 重排 $\tilde{F}_{2n}$ 的列, by 二进制反序

ex
$0 = 00_2 \to 0 = 00_2$, $1 = 01_2 \to 2 = 10_2$, $2 = 10_2 \to 1 = 01_2$, $3 = 11_2 \to 3 = 11_2$

ex $$\tilde{F}_4 = \begin{bmatrix} F_2 & \begin{bmatrix} 1 & \\ & -i \end{bmatrix} F_2 \\ F_2 & -\begin{bmatrix} 1 & \\ & -i \end{bmatrix} F_2 \end{bmatrix}$$

S_n eigenvalue: $\{\lambda_k\}_{k=0}^{n-1}$, $\lambda_k = e^{\frac{2\pi i k}{n}}$

$$\Rightarrow F_n^* S_n F_n = D_n = \begin{bmatrix} w^0 & & & \\ & w^1 & & \\ & & \ddots & \\ & & & w^{n-1} \end{bmatrix}$$

$$F_n^* (p(S_n)) F_n = p(D)$$

for $p(z) = \sum_{j=0}^{n-1} c_j z^j = c_0 + c_1 z + c_2 z^2 + \dots + c_{n-1} z^{n-1}$

Define: $P_{even}(z) := c_0 + c_2 z + \dots + c_{n-2} z^{\frac{n}{2}-1}$

$P_{odd}(z) := c_1 + c_3 z + \dots + c_{n-1} z^{\frac{n}{2}-1}$

$$\Rightarrow p(z) = P_{even}(z^2) + z P_{odd}(z^2)$$

$$= \frac{(1+z^{\frac{n}{2}})}{2} P_{even}(z) + \frac{(1-z^{\frac{n}{2}})}{2} P_{odd}(z/w_n)$$

ex $n=4$, $p(z) = a + bz + cz^2 + dz^3$

Given $p(1) = 2$, $\underset{\substack{\uparrow \\ w}}{p(i)} = 3$, $p(-1) = 5$, $p(-i) = 8$

求 a, b, c, d

Sol $$p(z) = \frac{1+z^2}{2} \overset{a+cz}{P_{even}(z)} + \frac{1+z^2}{2} \overset{b+dz}{P_{odd}(z/i)}$$

note, $P_{even}(z)$ 过 $(1,2)$ 和 $(-1,5)$ $\Rightarrow P_{even}(z) = \frac{1+z}{2} 2 + \frac{1-z}{2} 5 = \frac{7-3z}{2}$

同理, $P_{odd}(z) = \frac{1+z}{2} 3 + \frac{1-z}{2} 8 = \frac{11-5z}{2}$

$$\Rightarrow p(z) = \frac{1+z^2}{2} \frac{7-3z}{2} + \frac{1+z^2}{2} \frac{11-5(z/i)}{2}$$

$R = r_1 + 2r_2$

$F_2 \otimes F_3 \to F_6$ $\quad (0,0) \mapsto 0,\ (0,2) \mapsto 4,\ (1,1) \mapsto 3$

$(0,1) \mapsto 2,\ (1,0) \mapsto 1,\ (1,2) \mapsto 5$